AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 14 · STUDENT LESSON

Decision Trees

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Count the branch labels. Fit the tree. Freeze it before scoring.

Learn a one-question classifier under explicit majority, candidate-order and tie conventions.

Keep one habit: Training rows → candidate leaves and errors → saved tree → separate validation and baseline.

MAIN LESSON · SECTION 1

Which question and leaf predictions fit the training labels?

> Keep training-only fitting separate from validation predictions.

Your learning goals

I can fit a **root** question and **leaf prediction**s from **training** examples.

I can trace a new example through the learned tree.

I can compare a shallow tree with a baseline and state its limits.

Recall the need for separate validation

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Why does a cutoff with zero **training** errors still need validation on separate examples?

The tree you will fit

By the end, I can fit a one-question tree using **training** labels and an explicit candidate comparison.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

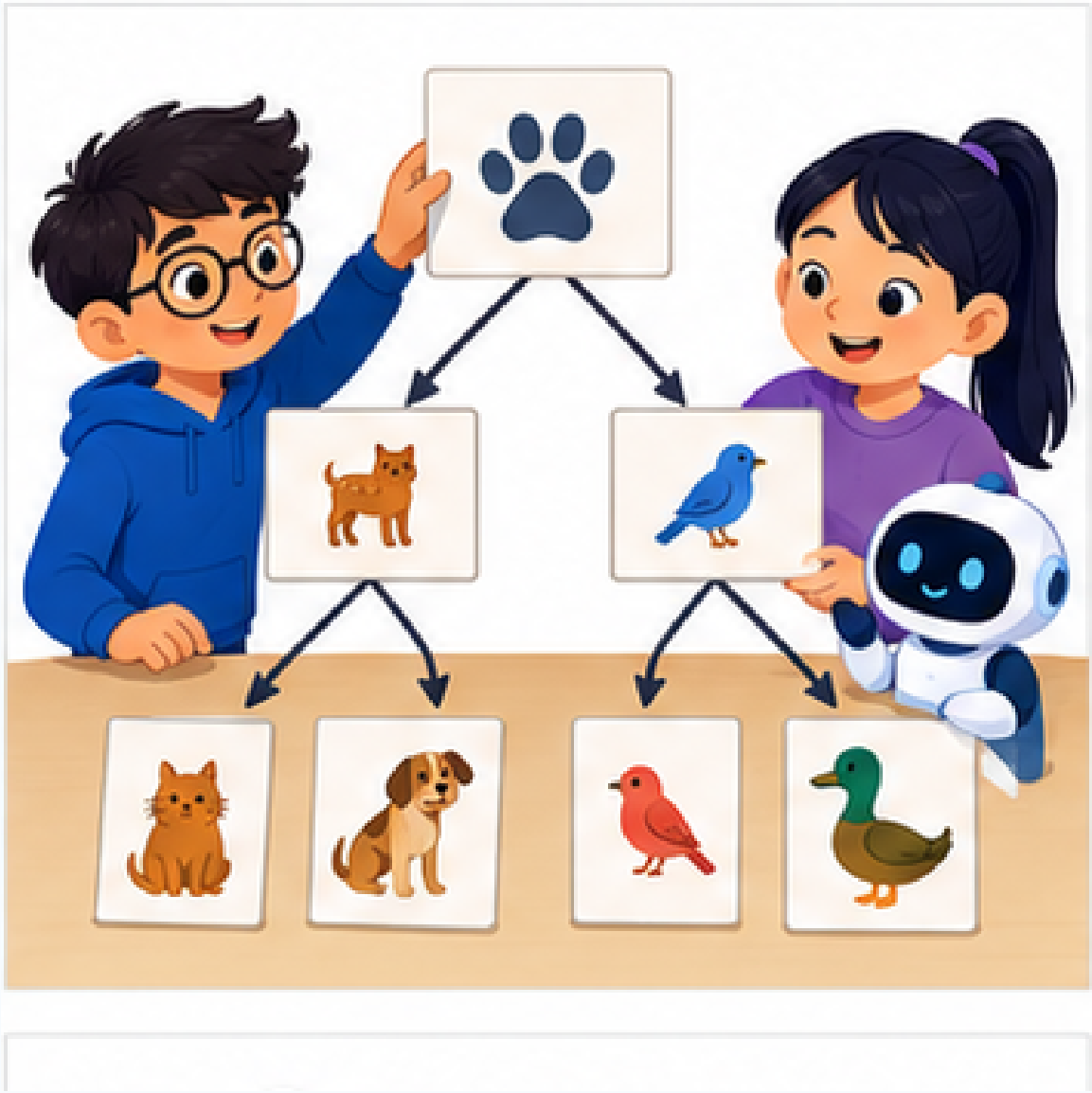
Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 14: **Decision Tree**s

Each question **partition**s possibilities; question order affects efficiency.



THINKER MISSION

Build a tree that identifies eight mystery animals.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Fit a one-question tree from six supplied training rows

> Both the selected flag and leaf predictions must be inspectable.

A root question, branches and leaves

A **decision tree** asks a feature question and follows a **branch** to a **prediction**. A tree with one question has **depth** one and is sometimes called a stump. In Week10 the Tower1 rule was supplied. Here the class names are still declared in advance, but training labels determine which question is selected and which class **prediction** is stored at each **leaf**.

The feature meanings and complete training table

Our six fictional **training** rows have two binary sensor flags available before **prediction**: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate.

Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

Row	Near	Tall	Label
1	1	0	blocked
2	1	1	blocked
3	1	0	blocked
4	0	1	clear

Row	Near	Tall	Label
5	0	0	clear
6	0	1	blocked

The candidate order, majority rule and tie conventions

Try near first and tall second. For each question, group the **training** rows by 0 and 1, then predict each group’s majority label; choose clear for tied label counts.

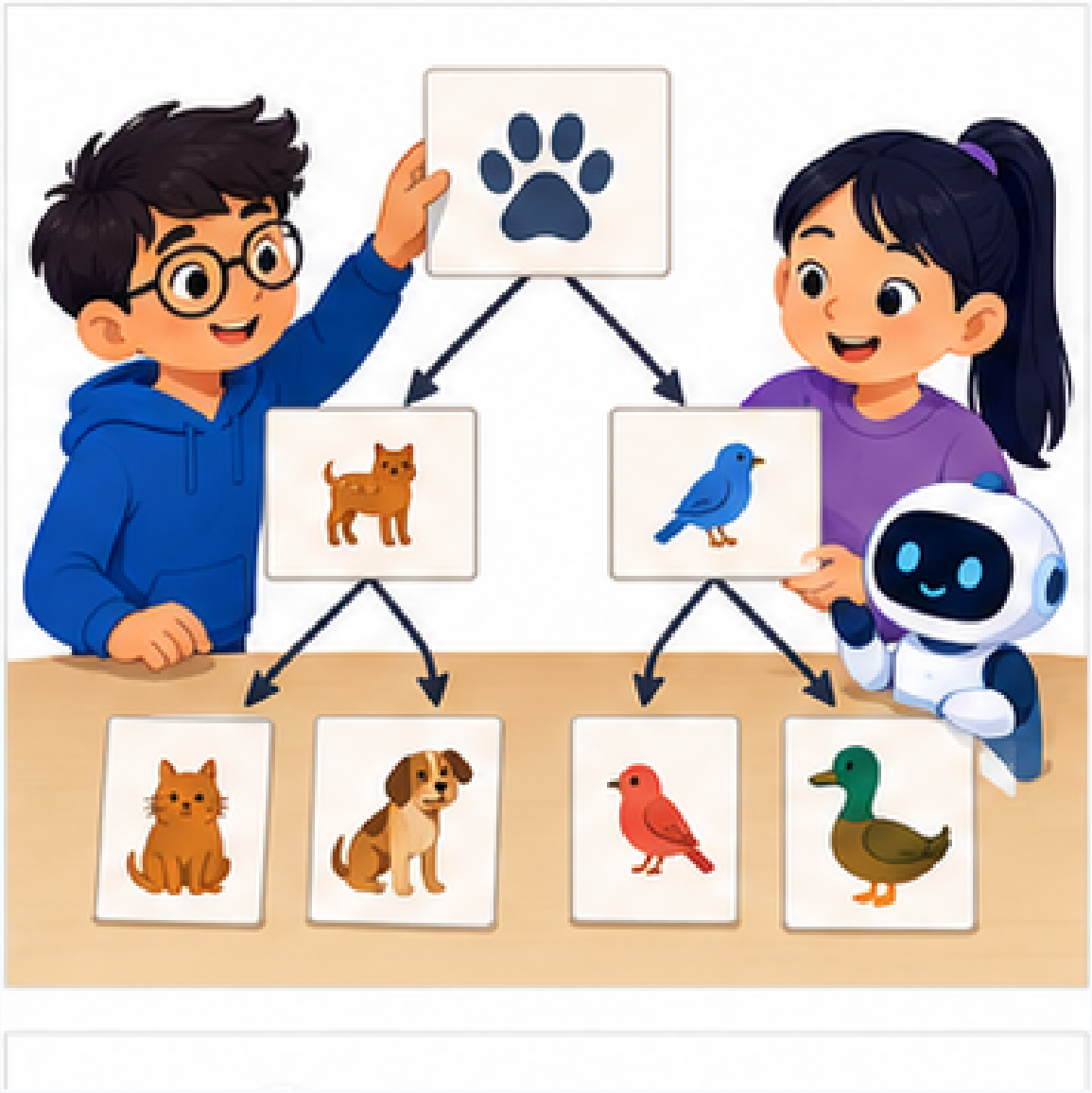
Count all **training** mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom **fitting** procedure.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **decision tree**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Read the complete depth-one fitting program

> This classroom learner minimises total training mistakes, not branch-size balance.

Understand the branch counts and strict comparison

Each row stores near at index0, tall at index1 and its reference label at index2. The inner list comprehension gathers training labels for one candidate's0 or1 branch. majority counts those labels and returns clear when counts tie. leaves maps each branch value to its predicted class. Score every training row once against its branch prediction. The strict < replacement means an equally scoring later question cannot replace the first. Fitting ends before any validation labels are used.

Predict and run both candidate comparisons

```
rows = [(1, 0, "blocked"), (1, 1, "blocked"),
        (1, 0, "blocked"), (0, 1, "clear"),
        (0, 0, "clear"), (0, 1, "blocked")]
def majority(labels):
    if labels.count("blocked") > labels.count("clear"):
        return "blocked"
    return "clear"
best_errors = len(rows) + 1
for feature in [0, 1]:
```

```
    leaves = {}
    for value in [0, 1]:
        labels = [row[2] for row in rows
                  if row[feature] == value]
        leaves[value] = majority(labels)
    errors = 0
    for row in rows:
        if leaves[row[feature]] != row[2]:
            errors = errors + 1
    print(feature, errors)
    if errors < best_errors:
        best_errors = errors
        tree = (feature, leaves)
print(tree[0], tree[1][0], tree[1][1])
```

Explain the selected near tree

Feature 0 is near; feature 1 is tall; row[2] is the label. **Output** is 0 1, then 1 2, then 0 clear blocked. Near makes one **training** mistake: near=0 contains two clear labels and one blocked, so that **leaf** predicts clear.

Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

Keep assumptions and the depth limit explicit

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both **branch**es of each candidate.

An empty **branch** would need a declared fallback; this small example does not implement one. The score is total **training** mistakes, not an average giving small and large **branch**es equal weight.

This is an explicitly limited **depth**-one learner. General tree algorithms can repeat split selection within **branch**es and often use impurity criteria such as Gini. Our code does not implement that full procedure.

Keeping the scope small lets you inspect every decision.

Point to a stored prediction

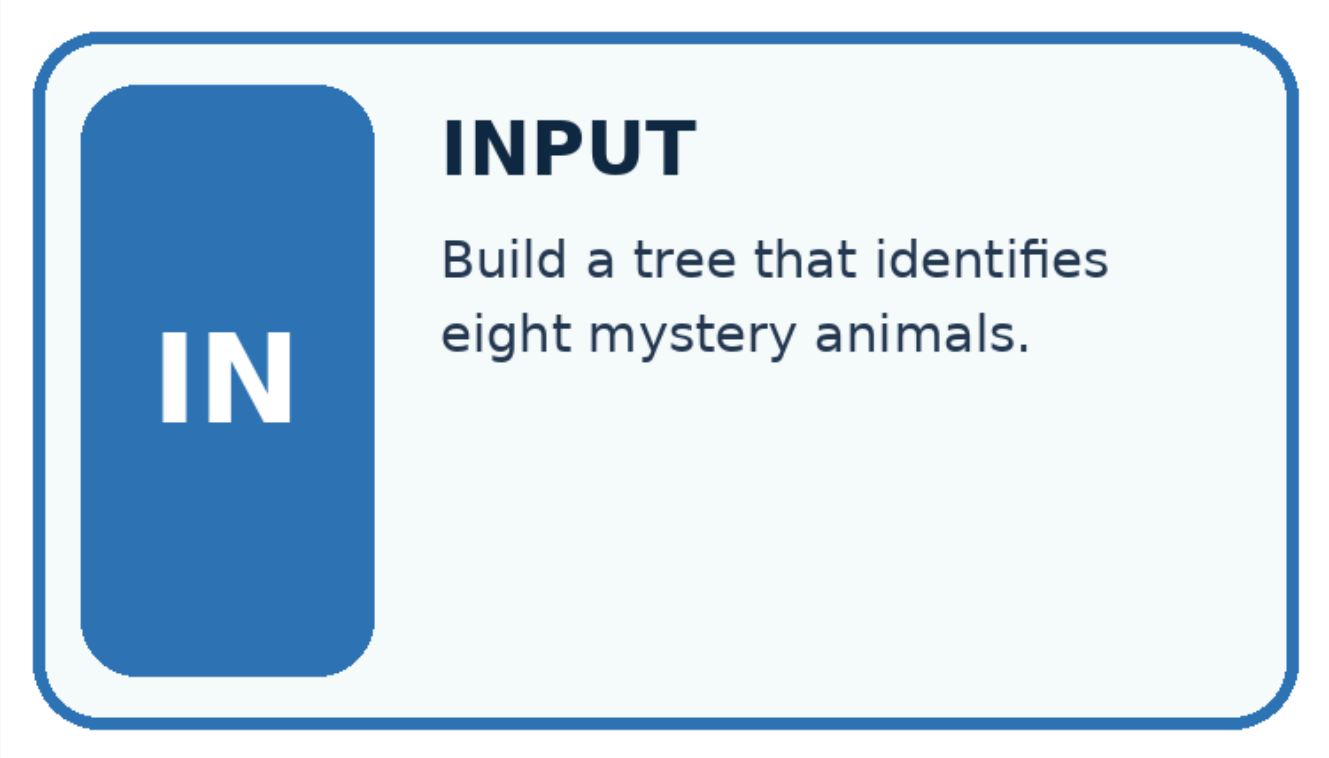
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

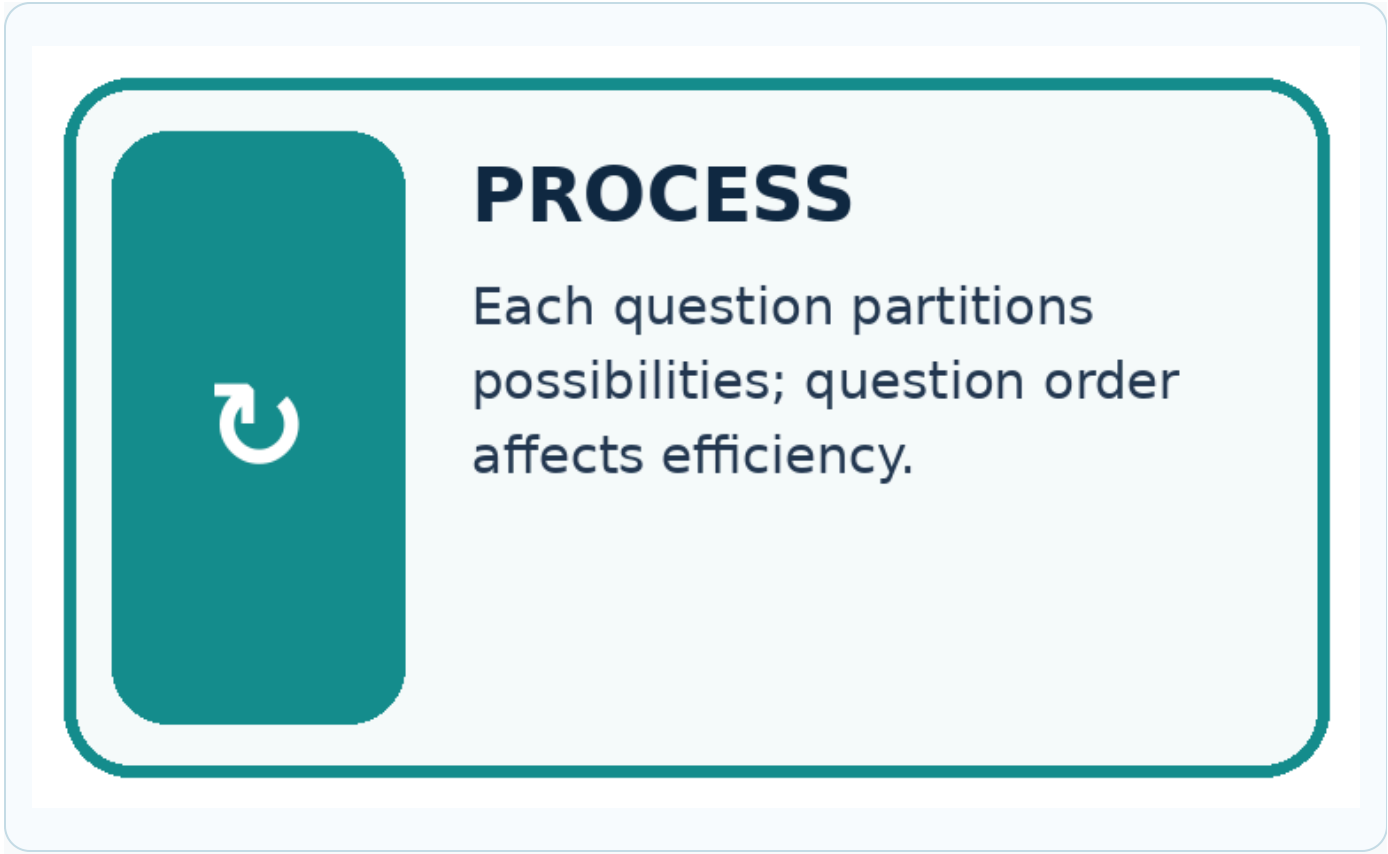
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
Each question partitions possibilities; question order affects efficiency.
↓
OUT
OUTPUT
Build a tree that identifies eight mystery animals.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

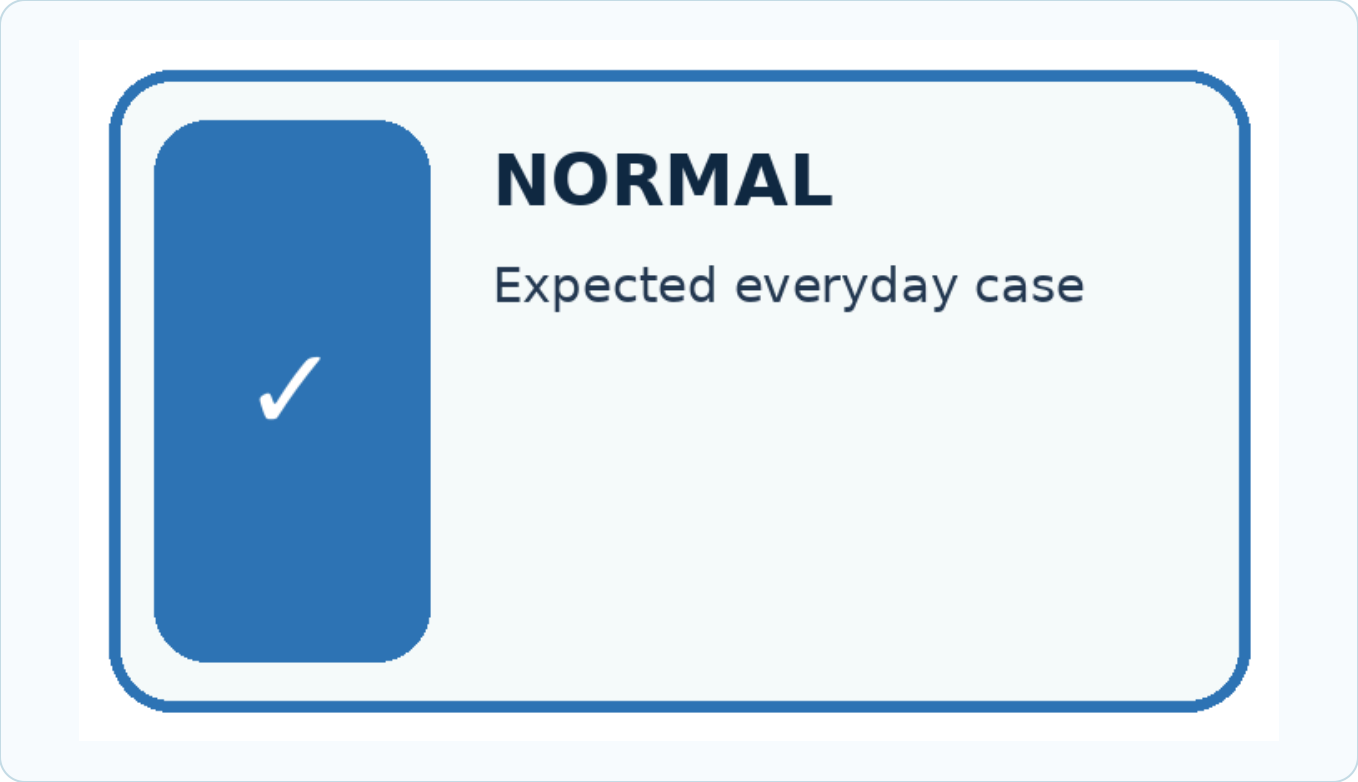
Connect the picture and the explanation

MAIN LESSON · SECTION 4

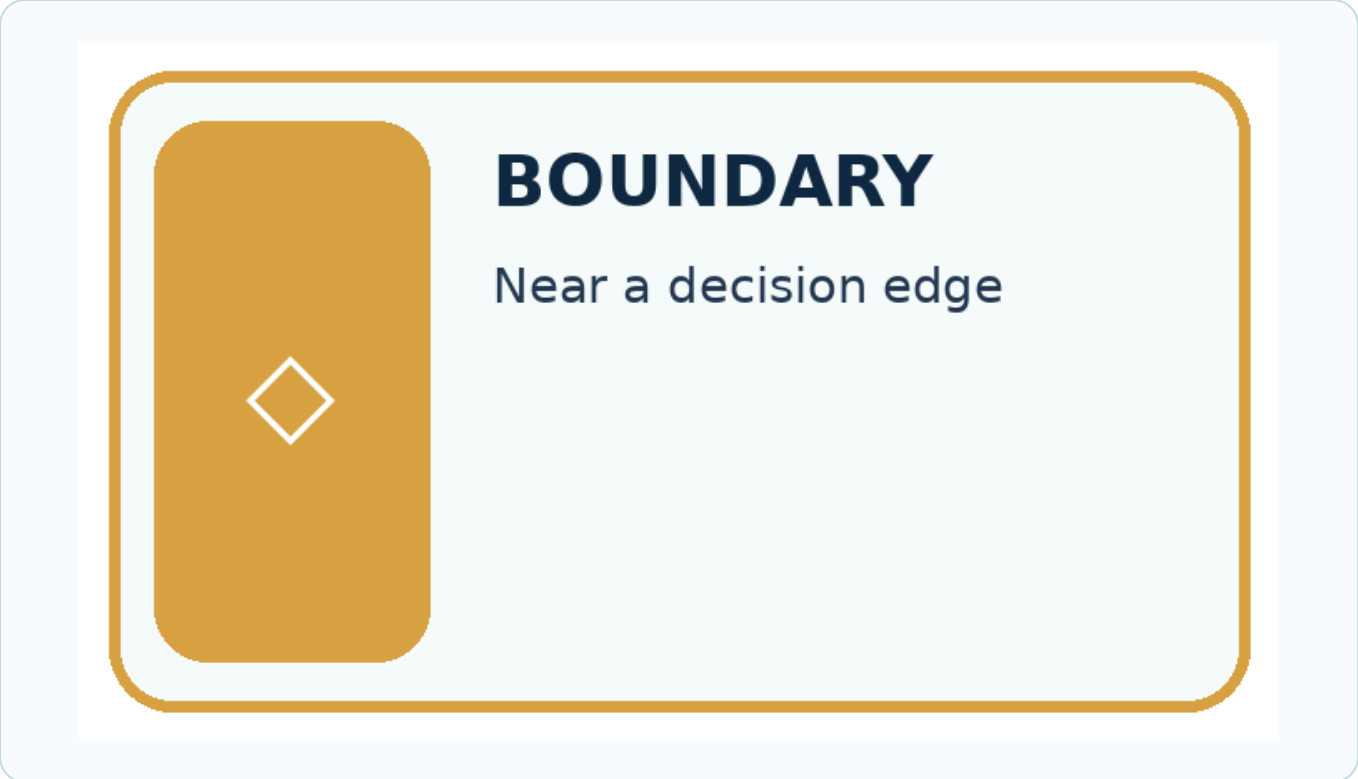
Challenge a fitted tree with a fresh trace

> A path follows the saved tree; it does not refit on each query.

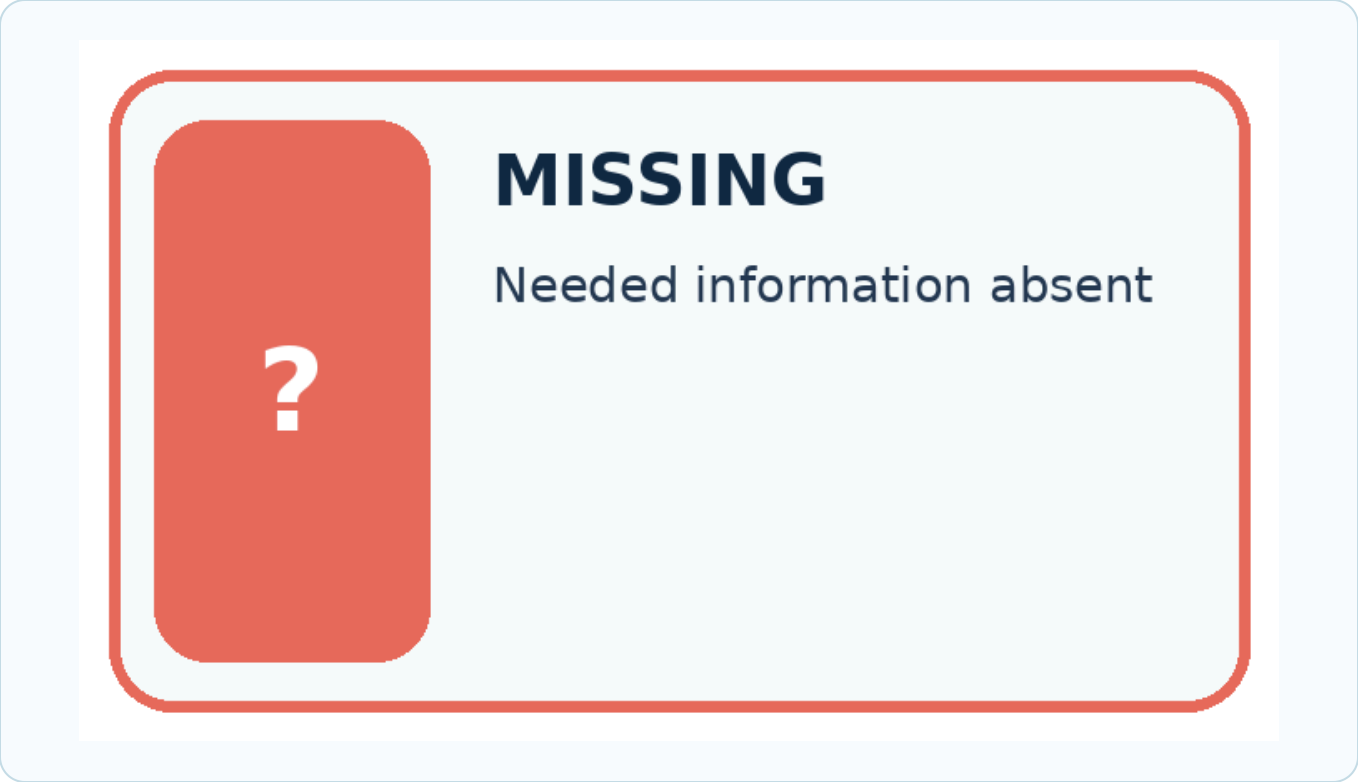
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: New **input** becomes (0,1).

▼ Check after predicting

Prediction: The fitted tree predicts Yes.

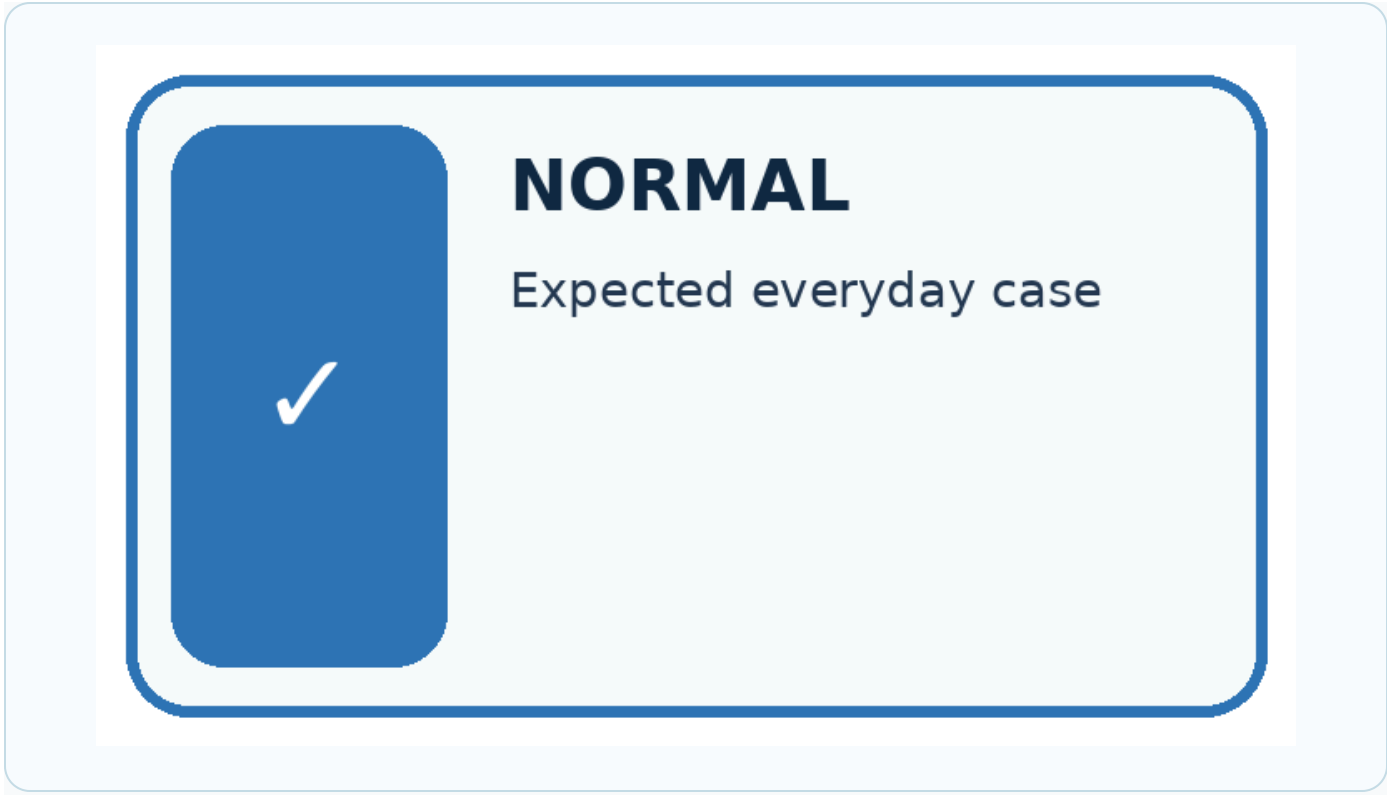
Reason: It follows diagram=1; cover colour is not the selected question.

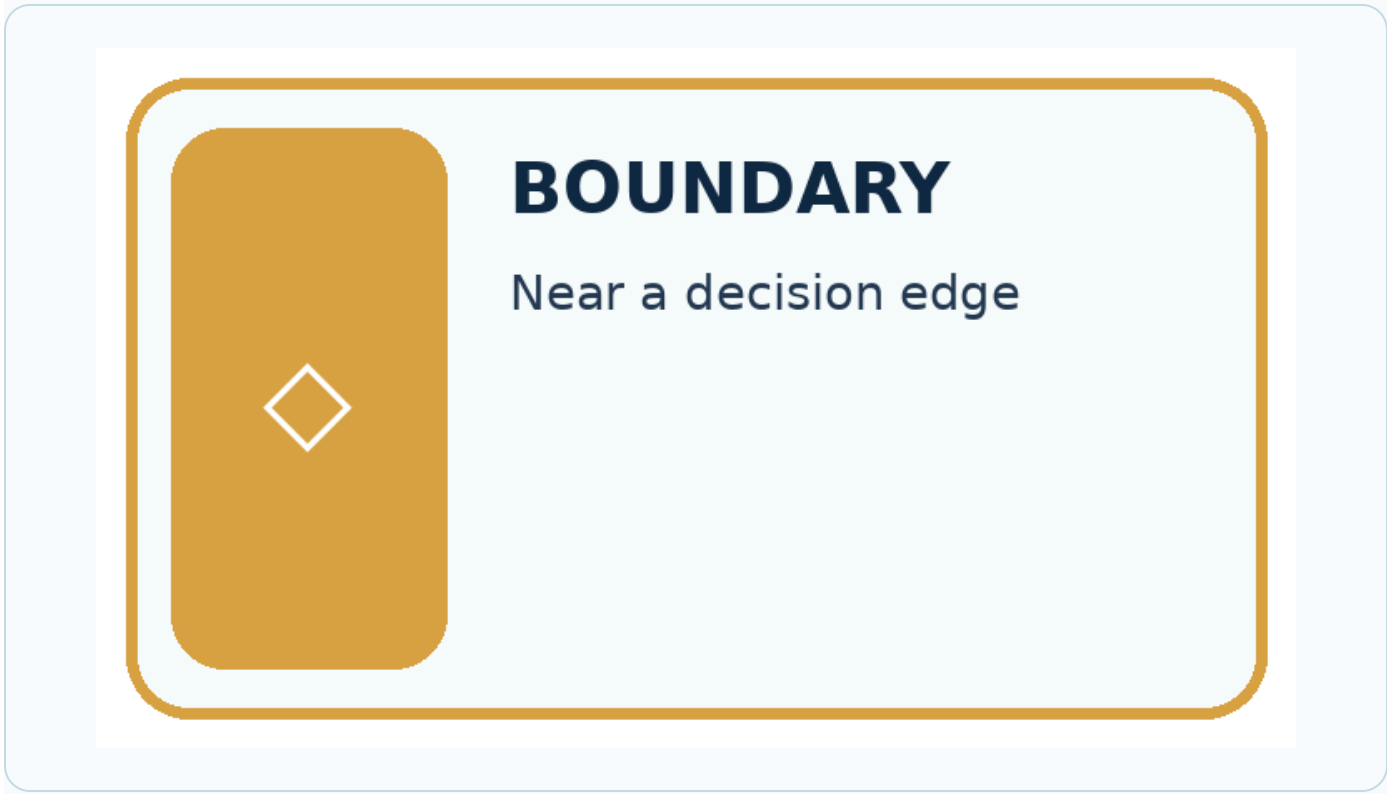
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

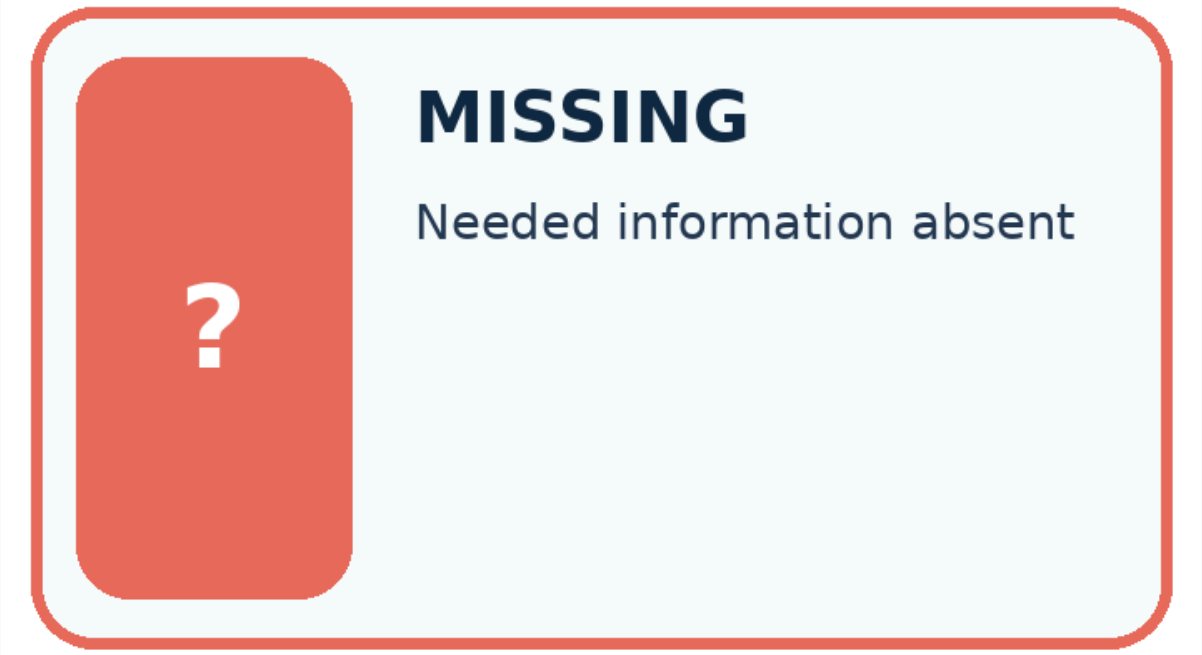
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Freeze the fitted tree, then score its validation predictions

> Keep the sensor tree distinct from the separate illustration-label tree.

Read the saved near decision

Two paths through the learned tree

ROOT QUESTION	BRANCH ANSWER	LEARNED LEAF
Is near equal to 1	No near is 0	Predict clear
Is near equal to 1	Yes near is 1	Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Predict the four validation outcomes

- Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: **prediction**s are blocked, clear, clear, blocked. Two of four match.

  The **training**-majority baseline predicts blocked, since **training** has four blocked and two clear; it also matches two of four validation labels.

Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.

Run the complete training-and-validation script

```
rows = [(1, 0, "blocked"), (1, 1, "blocked"),
        (1, 0, "blocked"), (0, 1, "clear"),
        (0, 0, "clear"), (0, 1, "blocked")]
def majority(labels):
    if labels.count("blocked") > labels.count("clear"):
        return "blocked"
    return "clear"
best_errors = len(rows) + 1
for feature in [0, 1]:
    leaves = {}
    for value in [0, 1]:
        labels = [row[2] for row in rows
                  if row[feature] == value]
        leaves[value] = majority(labels)
    errors = 0
    for row in rows:
        if leaves[row[feature]] != row[2]:
            errors = errors + 1
    print(feature, errors)
    if errors < best_errors:
        best_errors = errors
        tree = (feature, leaves)
print(tree[0], tree[1][0], tree[1][1])

validation = [(1, 1, "blocked"), (0, 0, "clear"),
              (0, 1, "blocked"), (1, 0, "clear")]
feature, leaves = tree
predictions = [leaves[row[feature]] for row in validation]
reference = [row[2] for row in validation]
print("tree", predictions, sum(p == y for p, y in zip(predictions, reference)),
len(reference))
```

```
baseline = majority([row[2] for row in rows])
print("baseline", baseline, sum(baseline == y for y in reference), len(reference))
```

Keep fitting and validation evidence separate

Keep the fitted **root** and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct **training prediction**s out of six is not the validation result.

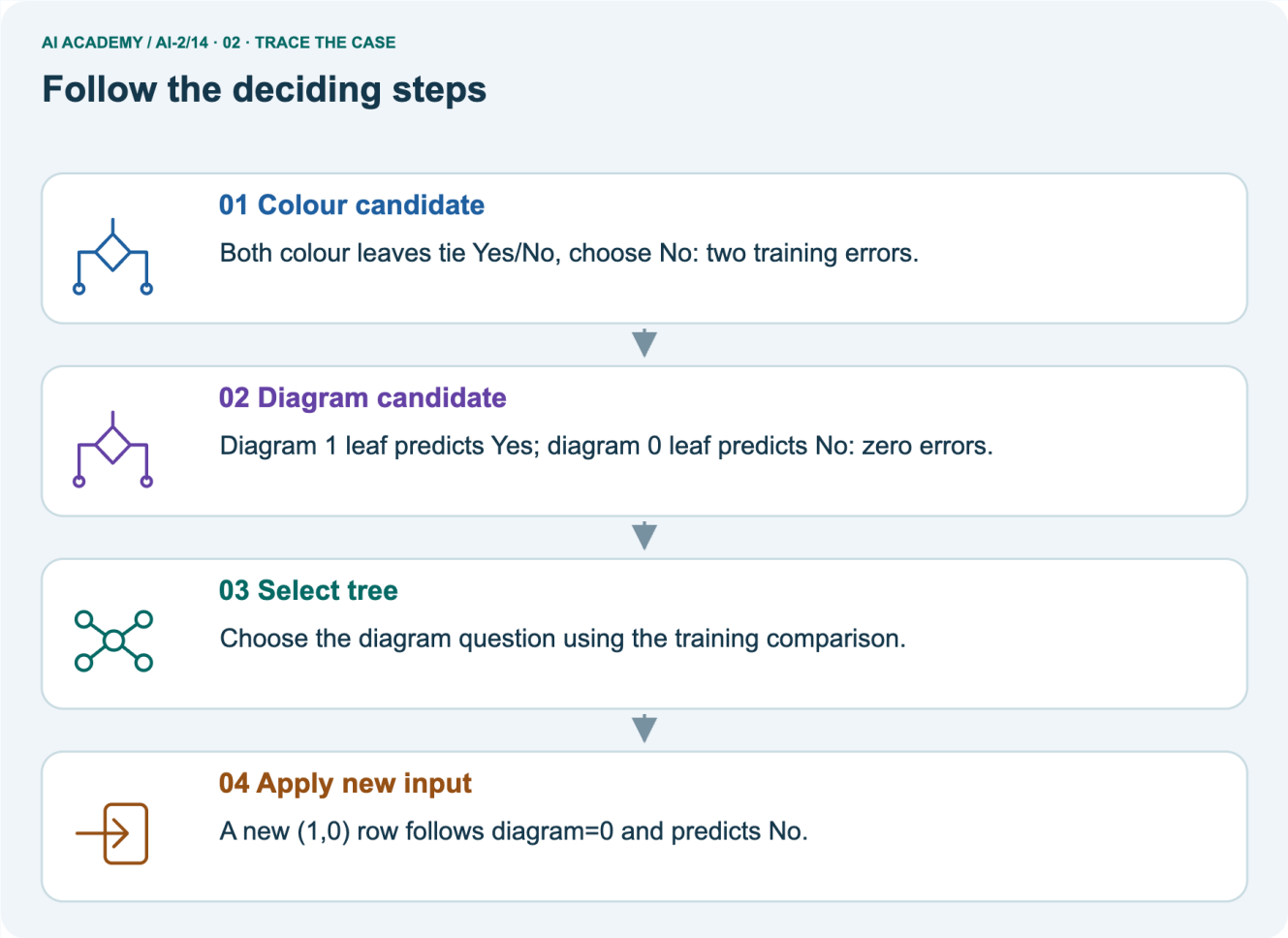
Introduce the separate four-book toy training set

A fictional toy classifier predicts whether a book needs an illustration label.

Training rows (colour_cover,diagram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No).

Candidates ask colour_cover or diagram_inside; each **leaf** uses its **training** majority. A **leaf** tie chooses No. Select fewer **training** errors.

Trace two candidates and their leaf counts



Follow the labelled picture with the matching explanation. Open the vector diagram

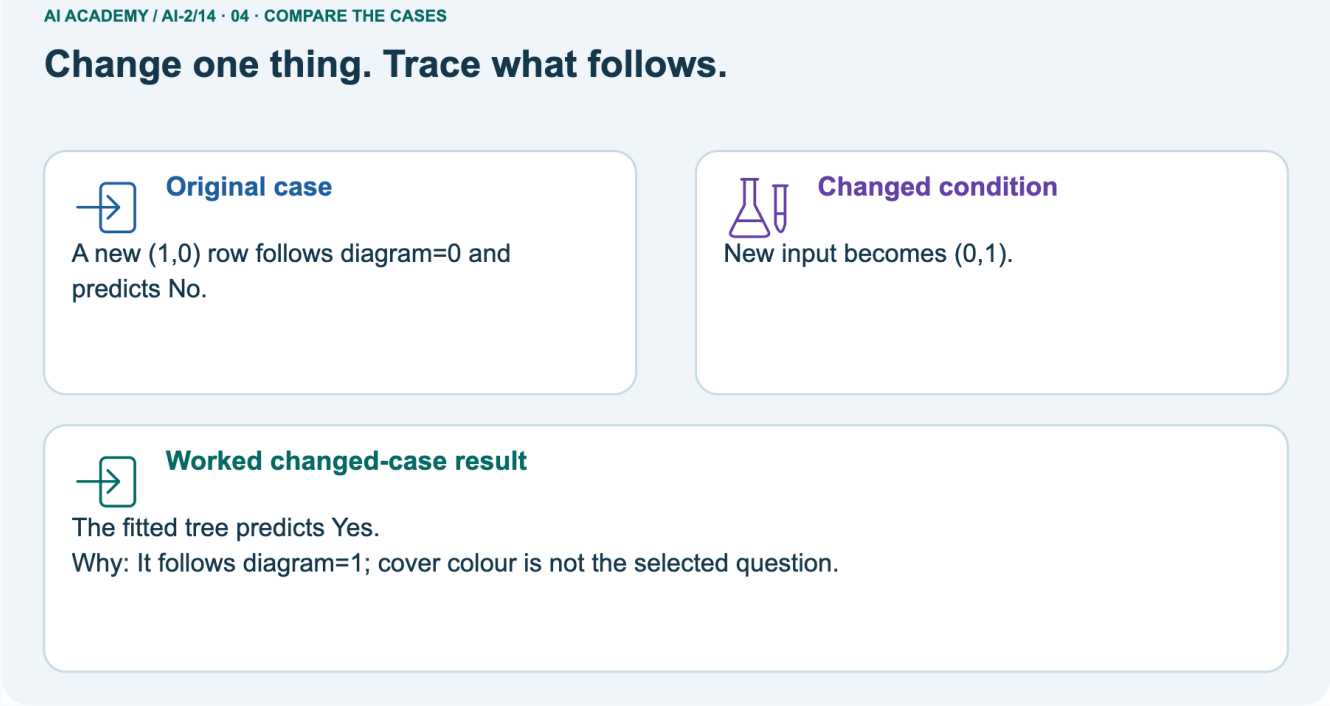
A question becomes a fitted tree only after its leaves and selection procedure are specified. The **training** comparison prefers the feature that separates these labels.

Prediction follows the selected question; it does not reselect the tree using a new case's answer. Save candidate leaves and errors so the choice can be repeated.

Predict a new diagram-flag input

- Change one thing: New **input** becomes (0,1). Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain feature dependence and the toy-data limit

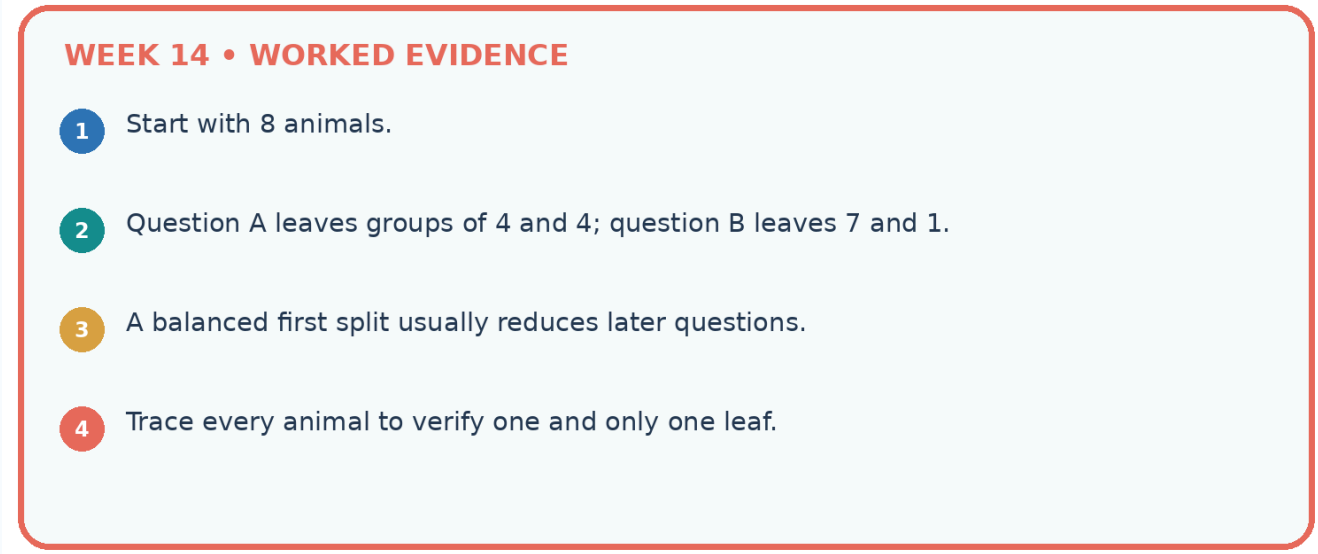
- Read the labelled picture: Why does changing colour alone not change this tree's **prediction**?
- Zero errors on four toy examples do not establish that diagrams always determine a real book's classification.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Start with 8 animals.

2. Question A leaves groups of 4 and 4; question B leaves 7 and 1.

3. A balanced first split usually reduces later questions.

4. Trace every animal to verify one and only one **leaf**.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

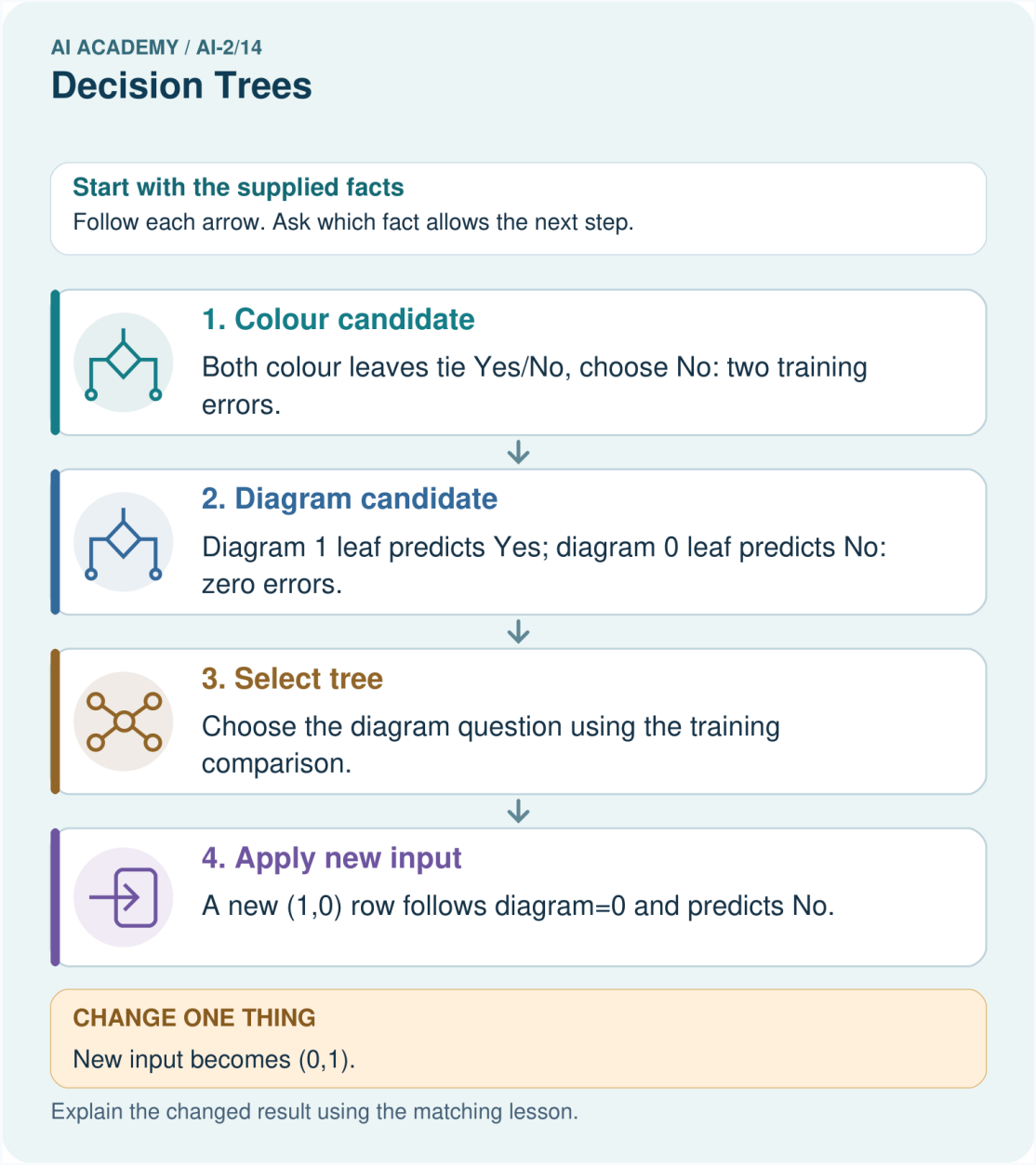
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Decision Trees

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Colour candidate	Both colour leaves tie Yes/No, choose No: two training errors.
2. Diagram candidate	Diagram 1 leaf predicts Yes; diagram 0 leaf predicts No: zero errors.
3. Select tree	Choose the diagram question using the training comparison.
4. Apply new input	A new (1,0) row follows diagram=0 and predicts No.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Keep the full fitting record and limit the reliability claim

> One pure training branch is not a guarantee about new trips.

Retain candidates, counts and versions

Record the two candidate questions, **branch** label counts, total errors, tie rules, **depth** limit, stored tree and validation **prediction**s.

Retain all rows, including the blocked example in the near=0 **branch**; removing a disagreement would change the **evidence**.

State the selected feature's limits

Our tree uses a single feature because that question scored better under this procedure. This neither establishes that near causes blockage nor that tall is useless in every dataset. Next is Week15, Choose Informative Questions. Carry forward candidate questions, branch label counts and training-only selection evidence when comparing how useful a question is.

Distinguish classification quality from a balanced partition

The earlier eight-animal illustration concerns how many questions an identification game might need. Equal-sized branches alone do not determine a good classification split: a four/four partition may mix both classes. This lesson instead compares the total training mistakes of declared majority leaves, with a depth limit and separate validation evidence.

Repair the reliability claim using validation

> **A learner says the tree is reliable because near=1 has only blocked training labels. Rewrite this claim using the leaf evidence, validation score and one limitation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A decision tree learns because arrows are drawn between boxes.
Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

The branch prediction comes from its labels, not its name

> A different training majority can reverse a leaf prediction.

Explain why near does not mean blocked

Do not assume near=1 must always predict blocked. If a new **training** set gives that **branch** a clear majority, this same learner will store clear there. A feature name is not the answer.

Use the **training** labels belonging to that **branch**.

Keep extra depth and final-test use explicit

Adding more questions can fit finer **training** details but does not guarantee better new-case performance. Record any **depth** choice made during development, and keep final test data separate.

A **leaf** majority describes this **training** group; it is not a promise about every future example.

Practise both candidate scores in W2

Use workbook W2 for the connected example "**Fitting** a **decision tree**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair the fixed-leaf misconception

- Use the worked case for **Fitting** a **decision tree**. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the separate diagram tree

Why does changing colour alone not change this tree’s **prediction**?

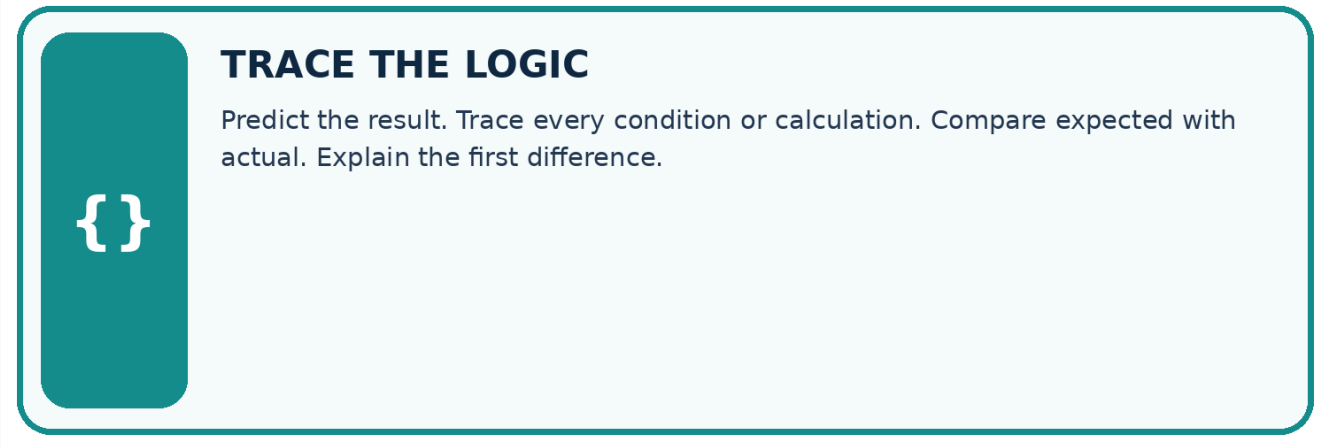
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Practise with less help: Verify the candidate scores

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Audit the supplied fitting and validation comparison

> Keep candidate fitting evidence and later predictions in separate records.

Predict, fit, freeze and compare

1. Use the six training rows in Section2 and four validation rows in Section6. Before execution, make four branch records: near0, near1, tall0, tall1. For each retain row IDs, clear/blocked counts, majority leaf and number of errors. Sum branch mistakes for each candidate; use near first, clear for tied leaf counts and the first candidate for tied scores. Save the chosen question and both leaves before examining validation results. Predict all four validation labels and the training-majority baseline; execute the complete comparison or label your work a paper trace. Preserve errors and both score denominators. Do not change a leaf using a validation label and still call it the original training-fitted tree. Reserve the new reversed-label training set for independent W4.

Use a tree-specific evidence record

Evidence	Record before moving on
Candidate branch	Question,0/1 branch and training row IDs
Leaf fitting	Clear count, blocked count, tie rule and chosen prediction
Candidate selection	Total mistakes, candidate order and strict comparison
Saved tree	Chosen feature and both stored leaves
Validation	All predictions, reference labels, matched IDs and count/4
Baseline and limits	Training-majority rule, same-case score and remaining uncertainty

Explain the revealing mismatch

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

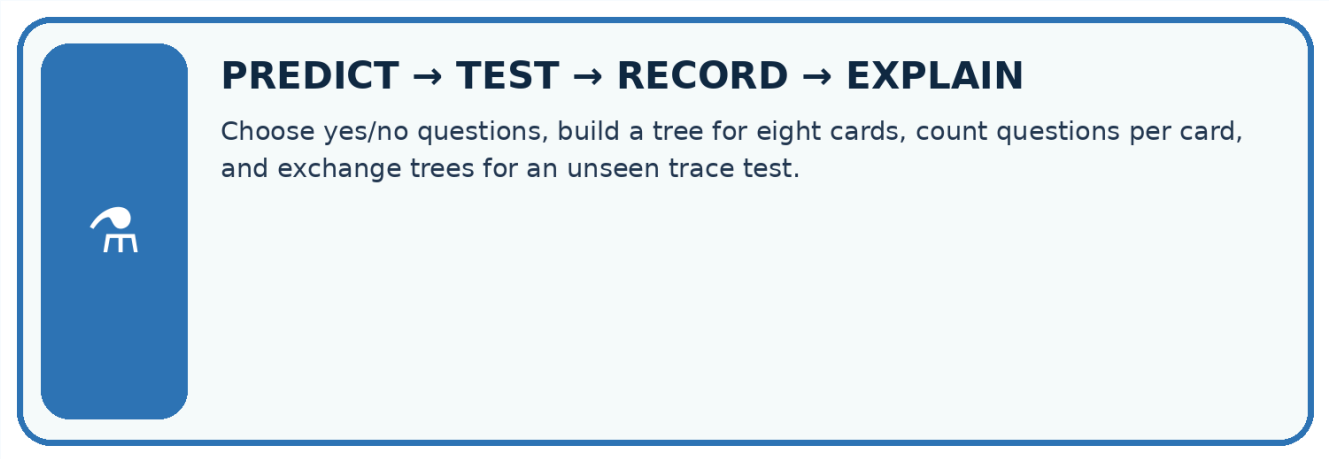
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the supplied six-row training set and four-row validation set. Compare the two candidate questions with their majority leaves, save the fitted depth-one tree and trace the separate validation cases. Keep the new independent training set reserved; no eight-card dataset is required.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: fit the tree to a different training set

> Use the declared policies rather than copying the worked leaf labels.

The independent training and validation records

New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree.

Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

The required candidate and tie policies

Fitting policy: candidates near first, tall second; flags use0/1. For each candidate and each branch count only its training labels, predict that branch's majority and choose clear for a tied leaf count. Add all training mistakes, replace the best candidate only for strictly fewer errors and stop at depth one. Freeze the selected question and both leaves for validation. Choose the baseline from the training-majority label, with clear for a tie.

Your first attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Fit a new tree independently

Week 14 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A tree asks topic is space, then pages at most 60. A space book has 70 pages. Trace both answers and explain why you cannot stop at the first question.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Create a transparent tree for hard constraints before applying preference scores.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Build a tree for science books: first ask science?; then, for yes, pages<=100? Trace science/100 and history/70. What should happen to science with missing pages?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Declare whether the recommender tree is specified or fitted

> Retain the appropriate source of each decision.

This week's inspectable portfolio contribution

Choose explicitly between two portfolio approaches. A hand-written hard-constraint tree follows a declared eligibility policy; it is not learned from examples. A fitted category tree uses candidate definitions and training labels to select its question and leaf predictions, then checks them on separate records. Name which approach you used, preserve the source policy or fitting record, and keep unresolved required inputs on a documented review path. Neither approach establishes reader enjoyment by itself.

Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and **training**-only selection **evidence**.

Keep the checkpoint, system and human choices visible

THIS WEEK'S CHECKPOINT Create a transparent tree for hard constraints before applying preference scores.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

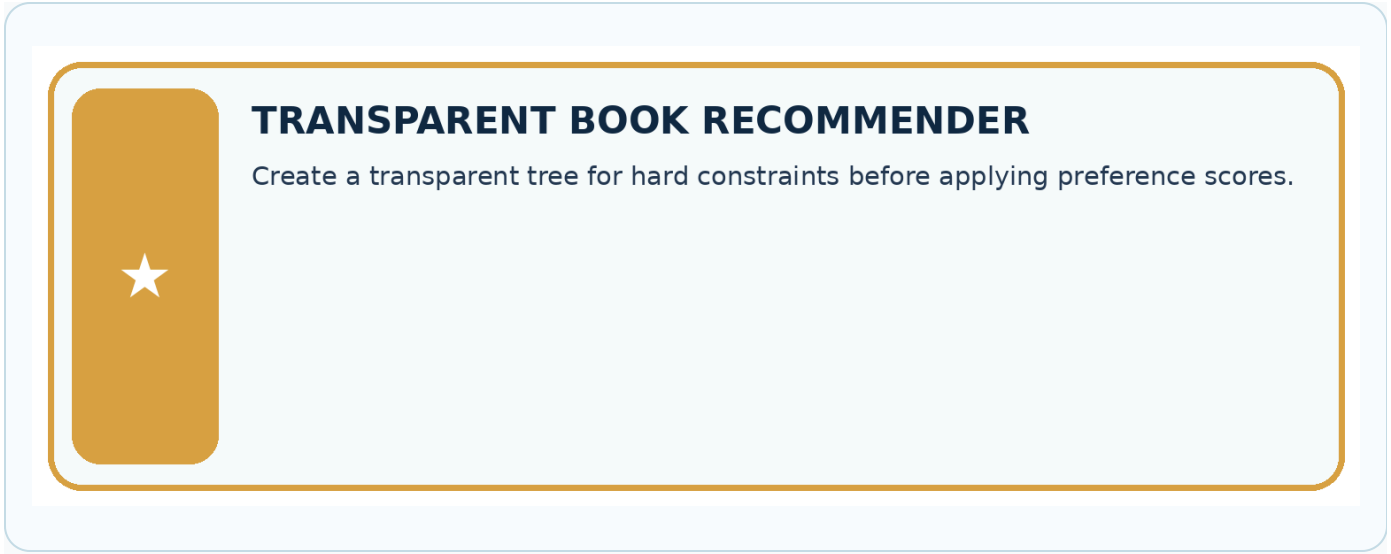
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain what was learned and what was only scored

> Name the training evidence, saved tree and separate validation result.

Your explanation without notes

- Close your notes. Explain this goal in your own words: Fit a **root** question and **leaf predictions** from **training** examples

My deciding step, evidence and one remaining question

Return on a changed case

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
decision tree	A sequence of branching questions that routes an input to an output at a leaf.	
node	A point in a graph or tree representing a state, question or result.	
leaf	A final node of a decision tree that supplies an output rather than another question.	
partition	One separated portion of a dataset, such as training, validation or test data.	

Terms for the connected explanation

Word	Meaning
Root	The first question asked by a tree

Word	Meaning
Branch	A path selected by the answer to a question
Depth	The largest number of question steps from root to leaf

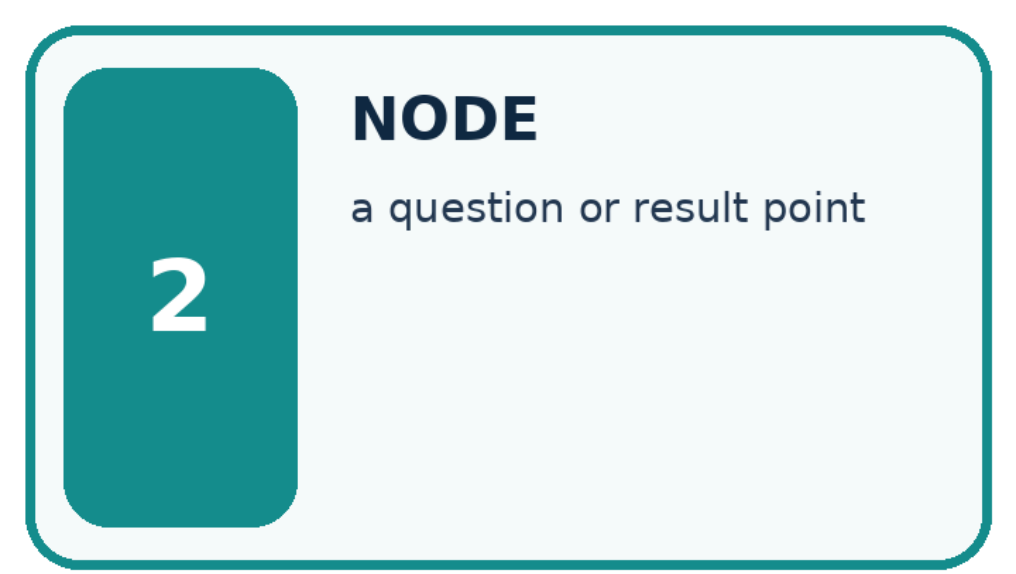
▼ Earlier wording, alternatives and picture transcripts

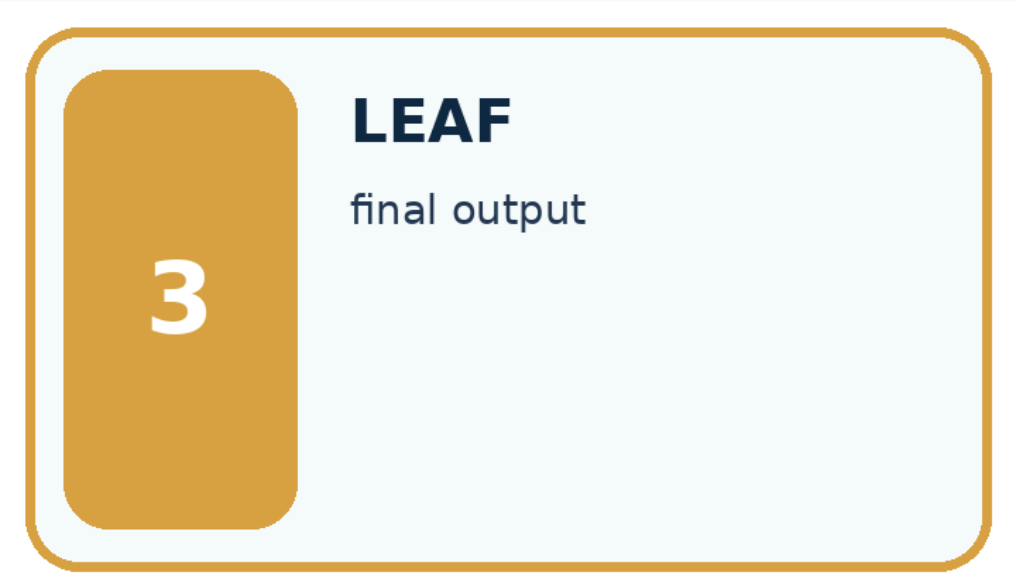
VOCABULARY LAB

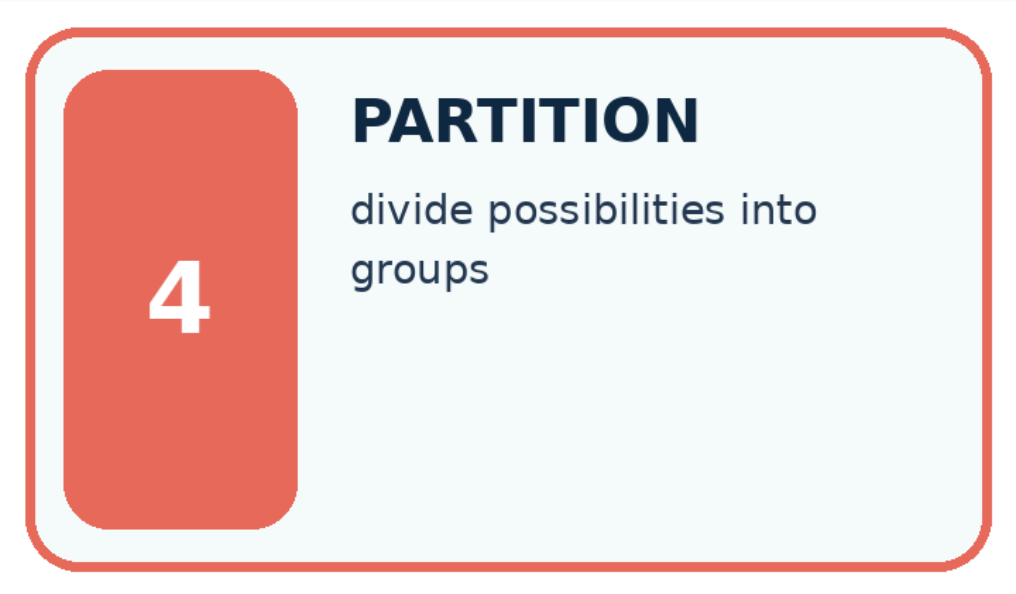
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
NODE
a question or result point


3
LEAF
final output



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, **branch**, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

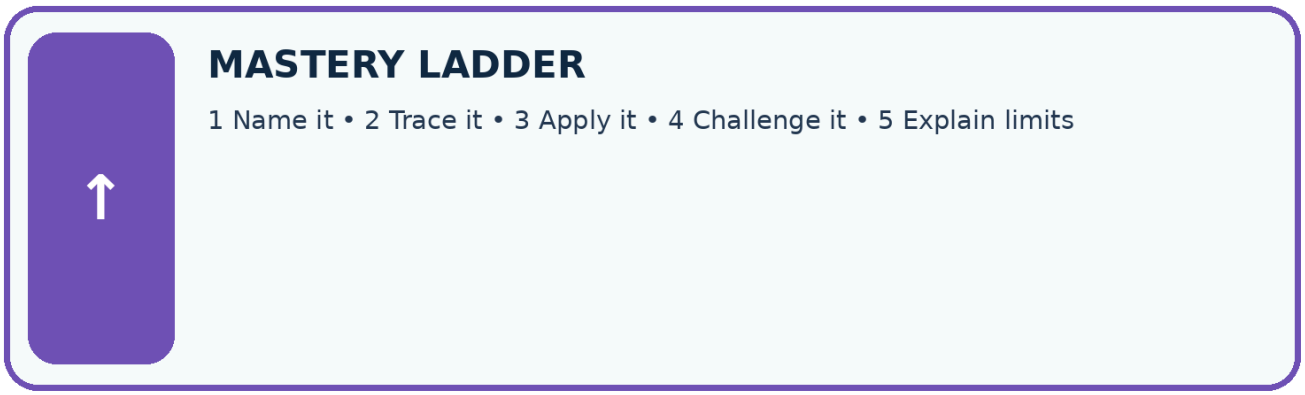
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Fit a new tree independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

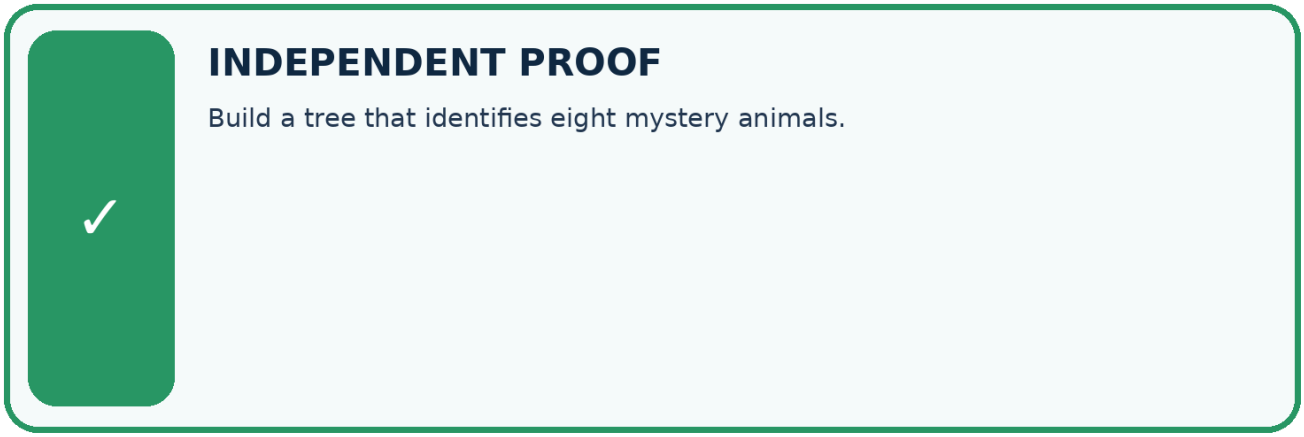
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Fit a new tree independently is the selected independent case.



EXPLANATION CHECK Fit candidate questions and leaf predictions from training labels, freeze the tree, then inspect separate validation evidence.

Write or draw your reasoning and evidence.

▼ **Extra practice: A decision tree can stop early**

REFERENCE · SECTION 13

Extra practice: A decision tree can stop early

> Try the question before opening its worked explanation.

Read the supplied case and question

First ask animal book? If no, Other. If yes, ask pages<100; yes gives Short Animal, no gives Longer Animal.

Trace 90-page animal and science books.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

The animal book reaches Short Animal; science reaches Other after the first question.

Evidence or stage	Value or result
Not an animal book	Other
Animal and fewer than 100 pages	Short Animal
Animal and at least 100 pages	Longer Animal

▼ Earlier wording, alternatives and picture transcripts

Week 14 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A decision tree can stop early

First ask animal book? If no, Other. If yes, ask pages<100; yes gives Short Animal, no gives Longer Animal.

Trace 90-page animal and science books.

The animal book reaches Short Animal; science reaches Other after the first question.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Fit a root question and leaf predictions from training examples	Explain the deciding step, show a trace or test, and state one limit.
Trace a new example through the learned tree	Explain the deciding step, show a trace or test, and state one limit.
Compare a shallow tree with a baseline and state its limits	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional toy classifier predicts whether a book needs an illustration label.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional toy classifier predicts whether a book needs an illustration label.
- Which labelled step matches this detail? A new (1,0) row follows diagram=0 and predicts No.
- 1. Select tree
- 2. Diagram candidate
- 3. Colour candidate
- 4. Apply new **input**

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Apply new **input** A new (1,0) row follows diagram=0 and predicts No. A question becomes a fitted tree only after its leaves and selection procedure are specified.

The **training** comparison prefers the feature that separates these labels. **Prediction** follows the selected question; it does not reselect the tree using a new case's answer.

Save candidate leaves and errors so the choice can be repeated.

Changed-case question

- Changed-case practice: New **input** becomes (0,1). Which conclusion is supported by the supplied facts?
- 1. The fitted tree predicts Yes.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The fitted tree predicts Yes. It follows diagram=1; cover colour is not the selected question.

Repair and save your work

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?.

Return to this week and explain what you changed.

Review **Training** and Testing (ai-2/11). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?.

Return to this week and explain what you changed.

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

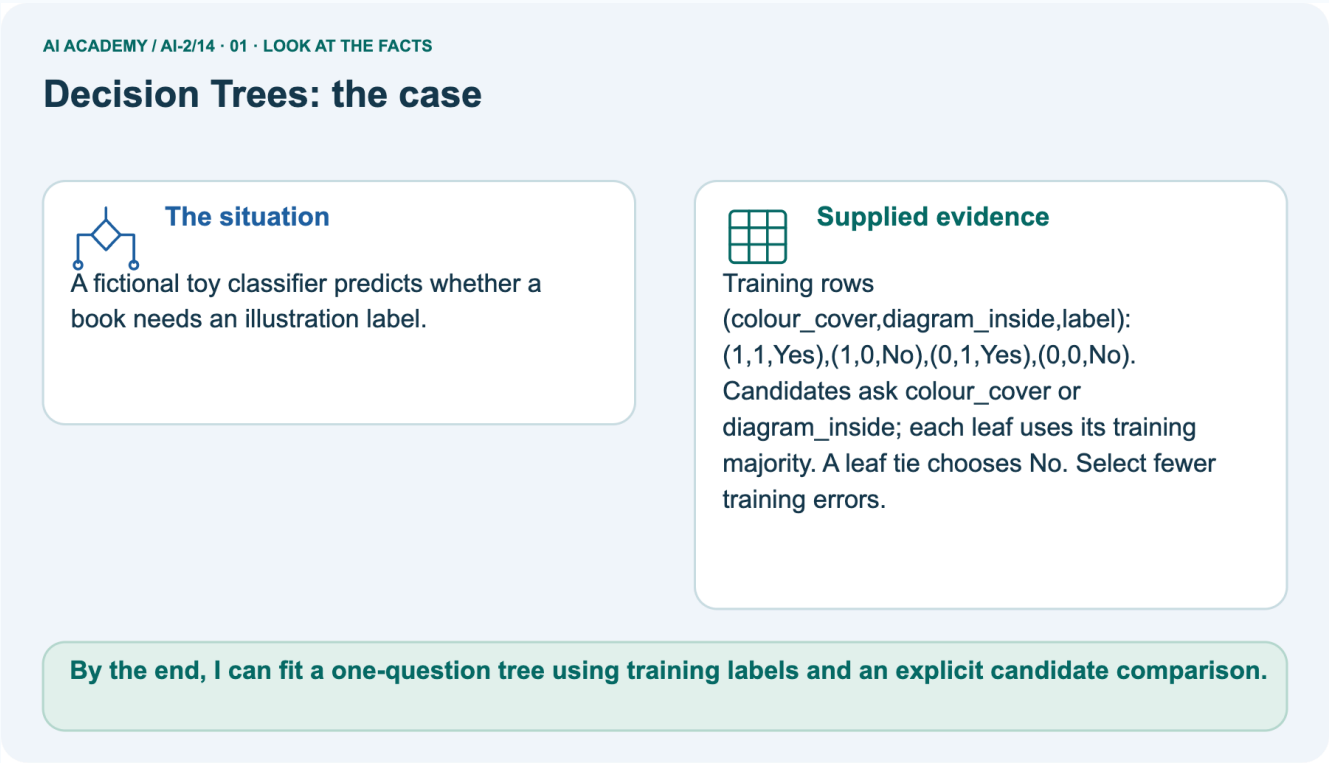
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Decision Trees: the case



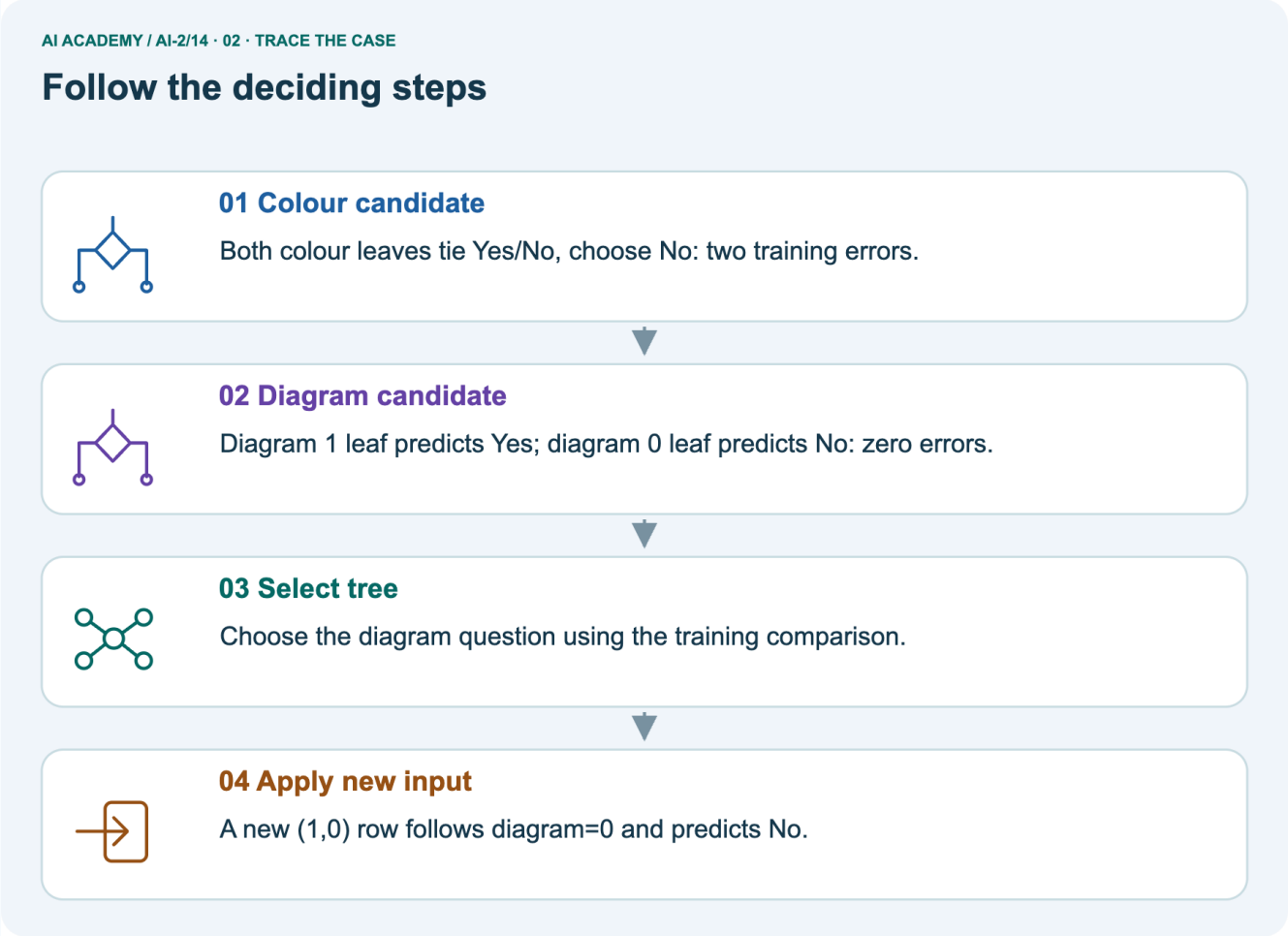
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional toy classifier predicts whether a book needs an illustration label. Supplied facts: **Training** rows (colour_cover,diagram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No).

Candidates ask colour_cover or diagram_inside; each **leaf** uses its **training** majority. A **leaf** tie chooses No. Select fewer **training** errors.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

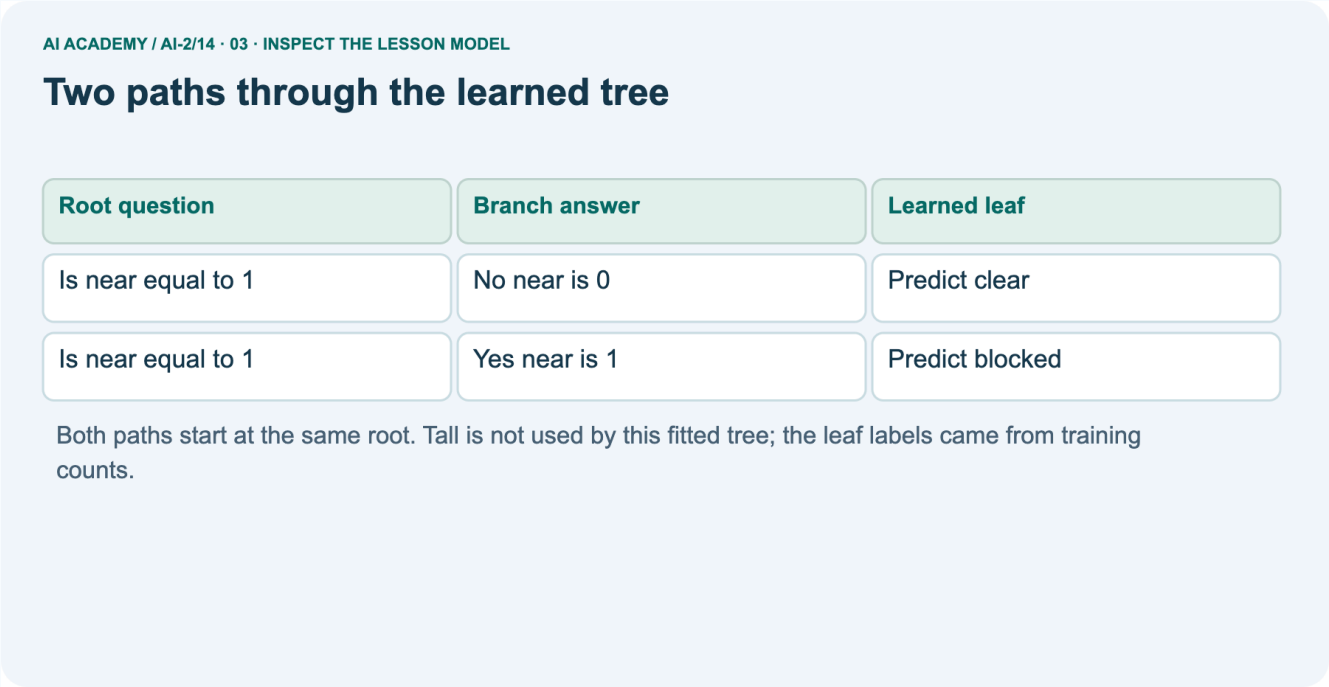


Follow the labelled picture with the matching explanation. Open the vector diagram

Colour candidate: Both colour leaves tie Yes/No, choose No: two **training** errors. → Diagram candidate: Diagram 1 **leaf** predicts Yes; diagram 0 **leaf** predicts No: zero errors. → Select tree: Choose the diagram question using the **training** comparison. → Apply new **input**: A new (1,0) row follows diagram=0 and predicts No.

Point to each numbered stage and explain which recorded **input** it uses.

03. Two paths through the learned tree

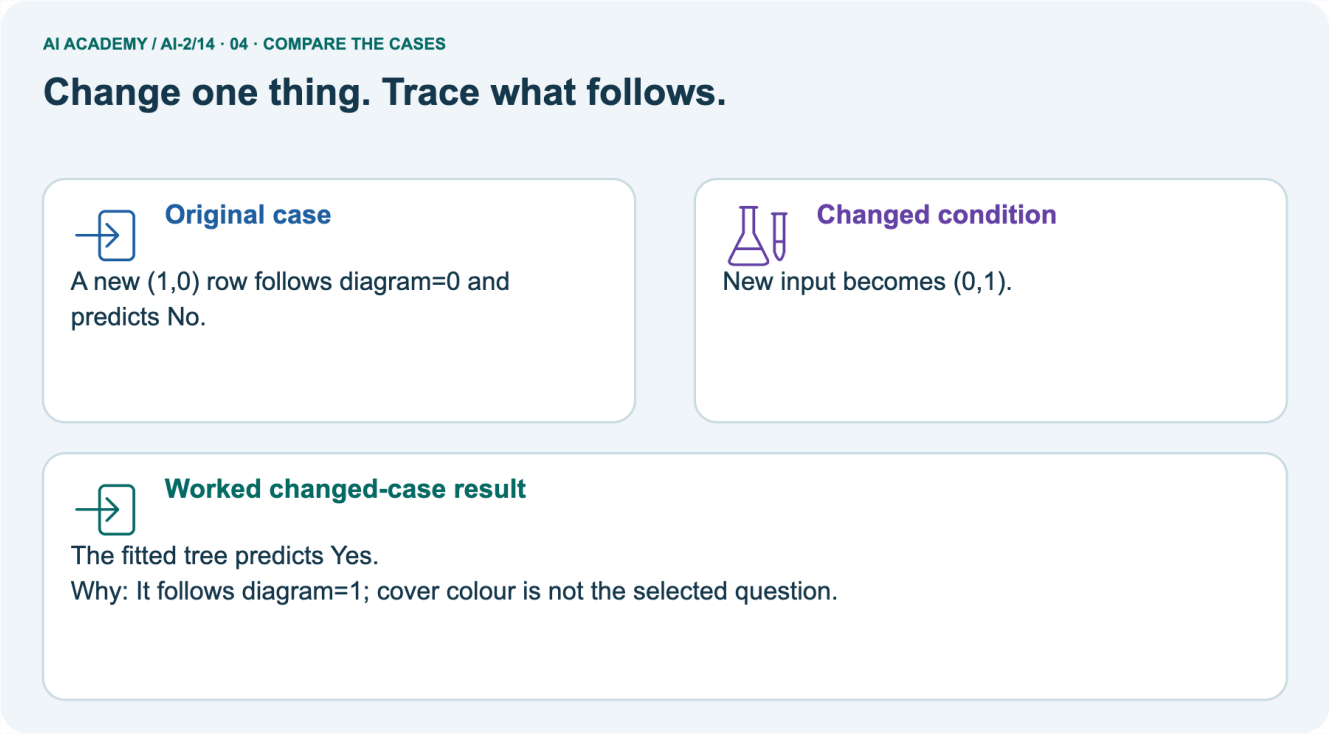


Follow the labelled picture with the matching explanation. Open the vector diagram

Two paths through the learned tree. Is near equal to 1 | No near is 0 | Predict clear. Is near equal to 1 | Yes near is 1 | Predict blocked

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



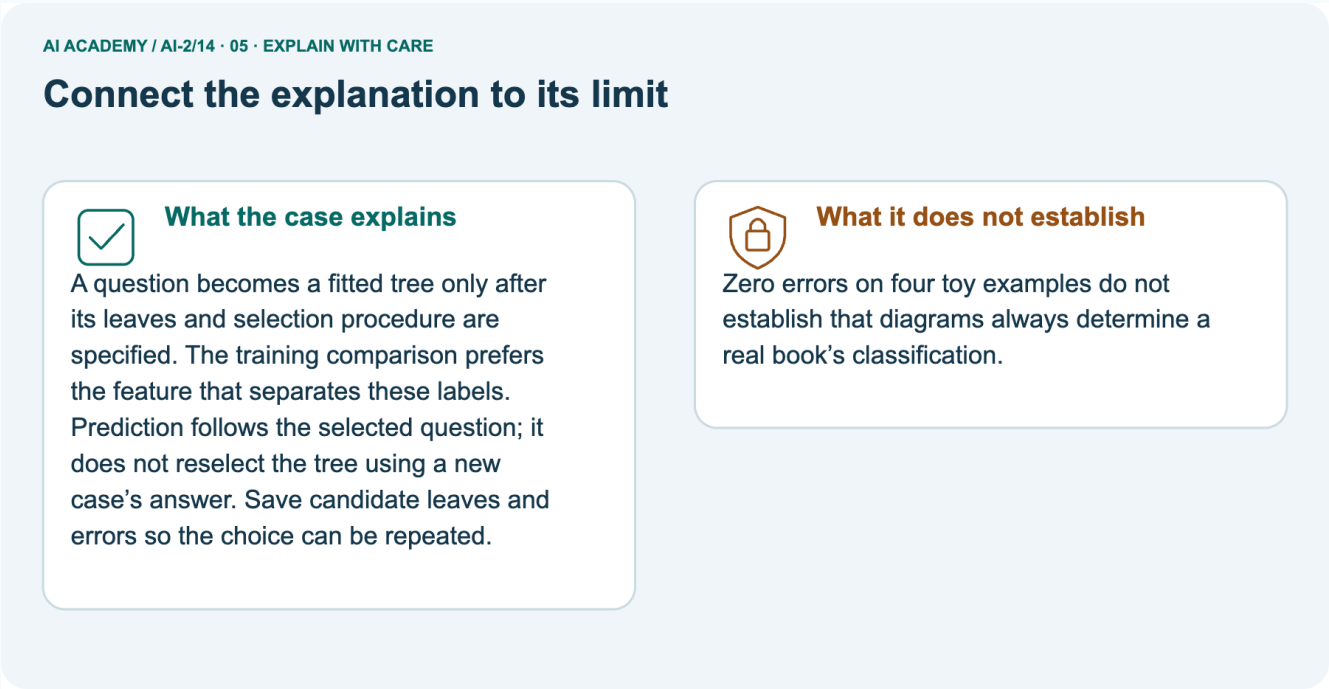
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A new (1,0) row follows diagram=0 and predicts No. Changed condition: New **input** becomes (0,1). Expected result: The fitted tree predicts Yes.

Reason: It follows diagram=1; cover colour is not the selected question.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

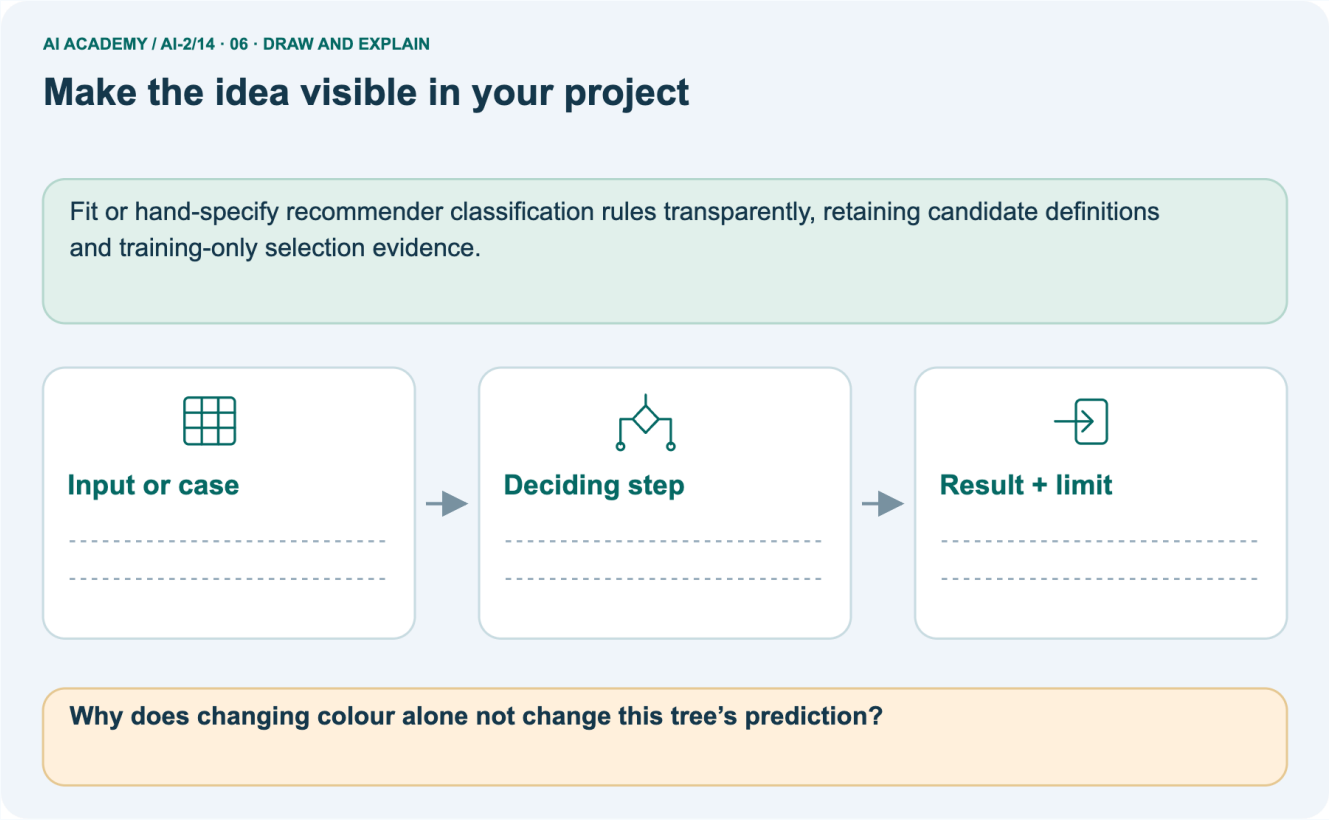
A question becomes a fitted tree only after its leaves and selection procedure are specified. The **training** comparison prefers the feature that separates these labels.

Prediction follows the selected question; it does not reselect the tree using a new case's answer. Save candidate leaves and errors so the choice can be repeated.

Limit: Zero errors on four toy examples do not establish that diagrams always determine a real book's classification.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and **training**-only selection **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Why does changing colour alone not change this tree's **prediction**?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Decision Trees: the case

02. Follow the deciding steps

03. Two paths through the learned tree

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.